

PRISM — Your campus, answered.

A conversational helper for placements and campus resources. One question. One clear answer with proof. One next step.

This document is the single source of truth for the research team. It contains the concept (in plain language), the core (how it works), the architecture (what we build in Databricks), and the research tasks to make it stronger before Sept 1.

SUPREME VISION

Edumerge-style OS (all campus data) + Personal Tutor-style adaptive coaching + Voice (Rumik/Cartesia) = PRISM Supreme. MVP for 12h is the center.

SUBMISSION STATUS

Idea submitted Aug 29-30 as PRISM (Track A). Selected for Hackathon Day Sept 1, 9AM-6PM, BMSCE (offline). Top 5 → Grand Finale.

1. Concept — In Layman’s Terms

Imagine a student in 3rd year. Infosys is coming next month. She wants to know: “Am I ready? What am I missing? Where do I fix it?” Today she runs between LMS, seniors, lab notices, and JD PDFs. The information exists, it’s just not connected.

PRISM is like that one friendly senior who knows the entire campus map — which skill needs which subject, which lab has the right setup, which seniors got into which company and how. But instead of finding that senior, you just ask PRISM like you’d ask a person.

“Am I ready for Infosys? Which lab can help me this week?” → “You’re missing Docker and System Design. Lab 3B is free tonight 7-10 with GPU, three seniors used it for the same company, here’s the course that covers it.”

For the placement team: *“Who needs help next month and why?”* — with real reasons, not just marks.

THE PERSONAL TUTOR TWIST

If you and your friend ask the same question, you get different answers because it knows *your* gaps. And you can say *“Test me on System Design”* — it asks you a question by voice (Rumik), you answer by speaking (Cartesia), and it gives feedback with the exact slide. Like a personal tutor sitting next to you, who can even speak in Kannada/Hindi.

WHY NOT JUST EDUMERGE?

Edumerge stores what happened. The personal tutor tells you what to study. PRISM tells you what to do *next, in your campus, tonight, with a voice that practices with you.*

2. Core — What Makes It Wow (Judging Lens)

Judges score 5 things for elimination. Here’s how PRISM maps:

JUDGING	WHAT WE SHOW	HOW WE PROVE
Usefulness	3 real users, daily pain, clear next step	Demo both views: student + placement officer
Innovation	Connected skill→course→lab→company map + voice	Graph + multilingual voice, not just chatbot
Genie at core	Genie is the experience, not an add-on	3 sharded Genies + Supervisor, live Genie function call
Feasibility	Sample data, fits Free Edition, 12h build	120k rows synthetic, no PII
Impact	Faster fixes, earlier help, less lab waste	Seconds not days, scalable to finale

Track A wants “complete journey: question → proof → action” — PRISM does exactly that.

3. Architecture — How We Build It (Databricks Native)

Medallion Lakehouse → Governance → Intelligence → Genie Core → Experience. This is CIDR 2021 Lakehouse pattern + Databricks 5th Genie Hackathon sharding pattern.

Sample Data (120k) → Lakeflow Designer → Delta Bronze/Silver/Gold (Unity Catalog)
→ Vector Search + Gold Skills Graph (skill→course→lab→company)
→ Agent Bricks Supervisor [Extraction | Retrieval | Forecasting]
→ 3× Genie Agents (Skills / Labs / Academic, each <10 tables)
→ Conversation API + AI/BI Dashboard overlay + Lakebase (live bookings)
→ MLflow 3.0 judges (groundedness)

LAYER 1: MEDALLION

Bronze raw → Silver cleaned ('ai_extract' JD→skills) → Gold business tables. Keep lineage for 10-sec lineage graph on stage.

LAYER 2: GOVERNANCE

Catalog 'prism' with comments/tags. UC Function 'prism.gold.is_lab_free(lab_id, ts)' — Genie calls it, doesn't hallucinate.

LAYER 3: GENIE CORE (3+1)

Don't make 1 big Genie (>30 tables fails). Shard into 3 narrow Genies + Supervisor App that routes and stitches. One chat for user.

LAYER 4: EXPERIENCE + OPS

AI/BI Dashboard with Genie overlay + Lakebase Postgres for live writes. Voice is thin layer: Cartesia STT → Genie → Rumik TTS.

TECH STACK (FREE EDITION)

Lakeflow Designer, Delta Lake, Unity Catalog, Lakebase, Vector Search, Agent Bricks (Multi-Agent Supervisor), Genie Spaces, AI/BI, Genie Code, MLflow 3.0, MCP. All free since June 2026.

4. Data — What We Use

All synthetic for demo, no personal data, fits Free Edition limits.

DATASET	ROWS	FIELDS	SOURCE
Students	5,000	dept, cgpa, attendance, skills_array	Synthetic via Assistant
Courses	40	name, skills_taught	Synthetic
Labs + Bookings	1,200	lab_id, gpu, free_start/end	Synthetic IoT logs
JDs	200	company, description, parsed skills	ai_extract
Placement history	3,000	student_id, company, year	Synthetic
City context	—	bus, internships	data.gov.in (optional)

Total ~120k rows. Generated via Assistant: “Generate 5k rows with schema ...” — 30 minutes. Bronze→Gold via Lakeflow.

5. Two Live Demo Questions (Must Rehearse 10x)

Q1 — Student view: “Am I ready for Infosys? Which lab tonight fixes my gap?” → Shows missing Docker/System Design, Lab 3B 7-10pm free, 3 seniors path, course slide. Then tap 📱 → “Mock me for System Design” → Rumik speaks, you answer, Genie evaluates.

Q2 — Officer view: “Show 25 students at risk next month and why.” → Shows at-risk list with reasons + suggested support. Follow-up: “What if they take Module 4 + Lab 3B?” → What-if simulation.

6. Name

Submitted as **PRISM** — keep it for Hackathon Day (judges have it). Team is Prometheus. For stage, introduce as **PYRA by Prometheus (PRISM)** — PYRA (pyr=fire, Greek, ties to Prometheus) or PHOS (light-bearer) for real product later. Don’t confuse sheet.

7. Research Tasks for Research Team (Before Sept 1, 9AM)

- **Validate 3 Genie golden SQLs** per agent (at-risk logic, lab free logic) against dummy data — ensure Genie returns cited tables.
- **Test Rumik TTS + Cartesia STT** on BMSCE wifi: latency, Kannada voice, fallback to text if fails.
- **Collect 5 real BMSCE lab names + 10 real JD skill lists** (Infosys, TCS) to make demo data feel real.
- **Prepare MLflow judge set:** 20 Qs with expected answer — for groundedness slide.
- **Print:** Architecture PNG A3 + 2 Qs script + lab list.
- **Do NOT build full data before 9AM** — only prompts/scripts. Full ingest happens on stage per rules.

References: CIDR 2021 Lakehouse (Zaharia), Databricks Docs: Unity Catalog / Lakeflow / Lakebase / Genie / Vector Search, Databricks Blog 5th Genie Hackathon (OneTrust sharding, ShipBob compose), SubgraphRAG 2410.20724 (graph+vector hybrid). All on Databricks Free Edition.

